

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

A Project Report Submitted in Partial Fulfillment of the Requirement for the Award of the Degree in

BACHELOR OF COMPUTER SCIENCE

Submitted By

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Declaration

I hereby declare that the project titled “**Streamlining ITProcurement: Automating Standard Laptop Orders with Flow Designer**” is an original academic project carried out as part of my degree programme. This project focuses on automating the Standard Laptop procurement process in ServiceNow using Flow Designer. The objective of the project is to streamline the request fulfillment process by automatically creating catalog tasks, updating task details, and assigning them to the appropriate hardware support team after approval.

I further declare that the information, implementation, and results presented in this project are based on my own academic work and understanding and have not been submitted previously to any other institution for the award of any degree or certification.

Abstract

The project “**Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer**” focuses on automating the standard laptop request process within the ServiceNow platform. In many organizations, IT procurement activities involve manual task creation, approval tracking, and assignment processes, which can increase processing time and administrative effort.

This project utilizes Flow Designer, a low-code automation tool in ServiceNow, to automate the workflow associated with Standard Laptop service requests. Once a user submits a laptop request through the Service Catalog and the request receives approval, the flow automatically creates a Catalog Task, updates the short description, and assigns the task to the Hardware support group.

The implementation demonstrates how workflow automation can improve operational efficiency, reduce manual intervention, ensure consistency in task management, and accelerate IT service delivery. The project also highlights the practical use of ServiceNow Flow Designer in automating procurement-related processes and enhancing overall user experience.

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1. INTRODUCTION

1.1 Project Overview

This project focuses on automating the Standard Laptop procurement process in ServiceNow using Flow Designer. In many organizations, laptop requests are handled manually, which increases processing time and administrative effort. This project automates the creation of catalog tasks after approval of a laptop request, ensuring faster and more efficient request fulfillment.

1.2 Purpose

The purpose of this project is to reduce manual intervention in IT procurement by automating the workflow for Standard Laptop requests. The automation ensures that approved requests are automatically assigned to the Hardware team with the required task details.

2. IDEATION PHASE

2.1 Problem Statement

In traditional IT procurement systems, laptop requests require manual task creation and assignment after approval. This process is time-consuming and may lead to delays or errors.

2.2 Empathy Map Canvas

Users:

- Employees requesting laptops
- Managers approving requests
- Hardware support team

User Needs:

- Quick request processing
- Automated task creation
- Faster laptop delivery

2.3 Brainstorming

Possible solutions considered:

- Manual processing
- Email-based approval
- ServiceNow workflow automation

The Flow Designer automation was selected because it provides faster processing and reduces human effort.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. User opens Service Catalog.
2. User selects Standard Laptop.
3. User places order.
4. Manager approves request.
5. Flow automatically creates Catalog Task.
6. Hardware team receives task.
7. Laptop is configured and delivered.

3.2 Solution Requirement

Functional Requirements

- Create Flow in Flow Designer , Update short description.
- Trigger Flow through Service Catalog request.
- Create Catalog Task automatically , Assign task to H/W groups.

Non-Functional Requirements

- Reliability
- Automation
- Accuracy
- Faster processing

3.3 Data Flow Diagram

User → Service Catalog → Approval → Flow Designer → Catalog Task
→ Hardware Team

3.4 Technology Stack

Technology	Purpose
Servicenow	Platform
Flow Designer	Automation
Service Catalog	Request Management
Catalog Task	Task Creation

4. PROJECT DESIGN

4.1 Problem Solution Fit

The proposed solution eliminates manual task creation by automatically generating catalog tasks after approval.

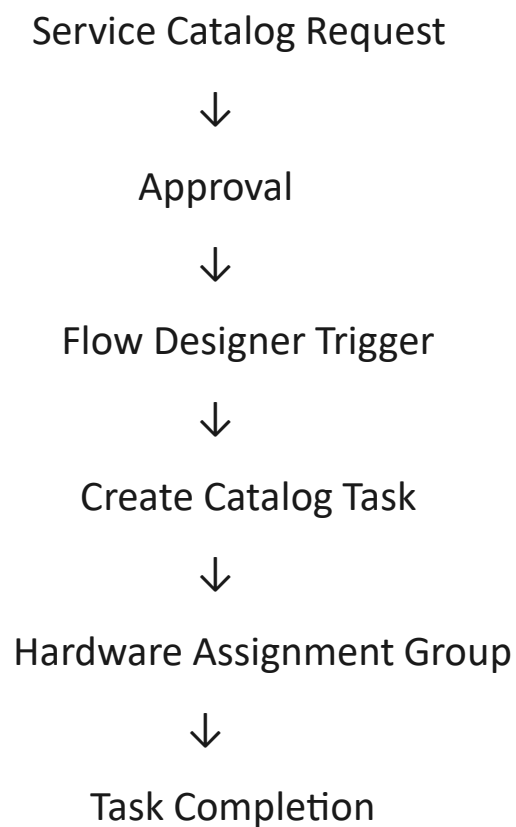
4.2 Proposed Solution

A Flow Designer workflow is created that:

- Monitors Standard Laptop requests.
- Waits for approval.
- Creates Catalog Task.
- Updates description.
- Assigns Hardware group.

4.3 Solution Architecture

Flow Process:



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Milestone 1 : *Flow*

Create a Flow to Create Catalog task to update the short description after Approval the service Request and it will be Assignment group to Hardware.

Figure 5.1: Searching Flow Designer in ServiceNow

This screenshot shows searching and opening Flow Designer from the ServiceNow navigation menu.

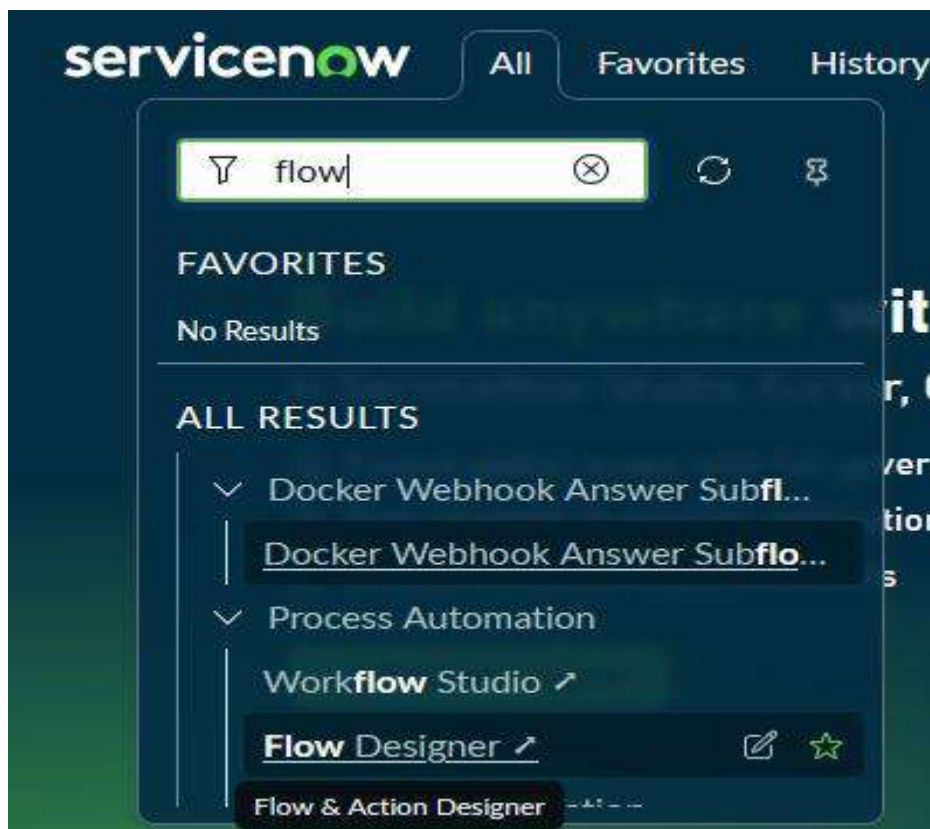


Figure 5.2 Creating a New Flow

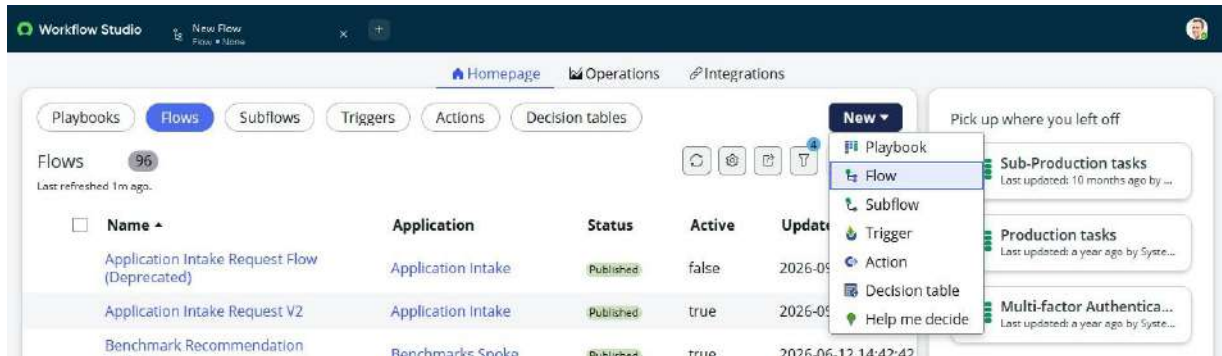


Figure 5.3: Flow Properties Configuration

The flow properties such as name, application, and run-as user are configured.

Let's get the details for your flow

Flow name * ⓘ

Application * ⓘ

Description ⓘ

▼ **Hide additional properties**

Protection ⓘ

Run as ⓘ

Flow priority default ⓘ

Cancel

Build flow

Figure 5.4: Selecting Service Catalog Trigger

The Service Catalog trigger is selected to start the flow automatically.

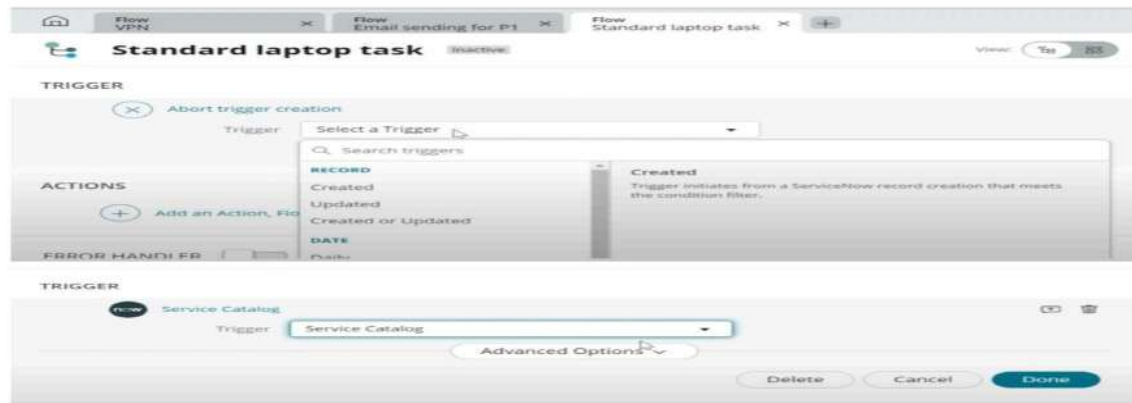


Figure 5.5: Adding Create Catalog Task Action

The Create Catalog Task action is added to automate task creation.

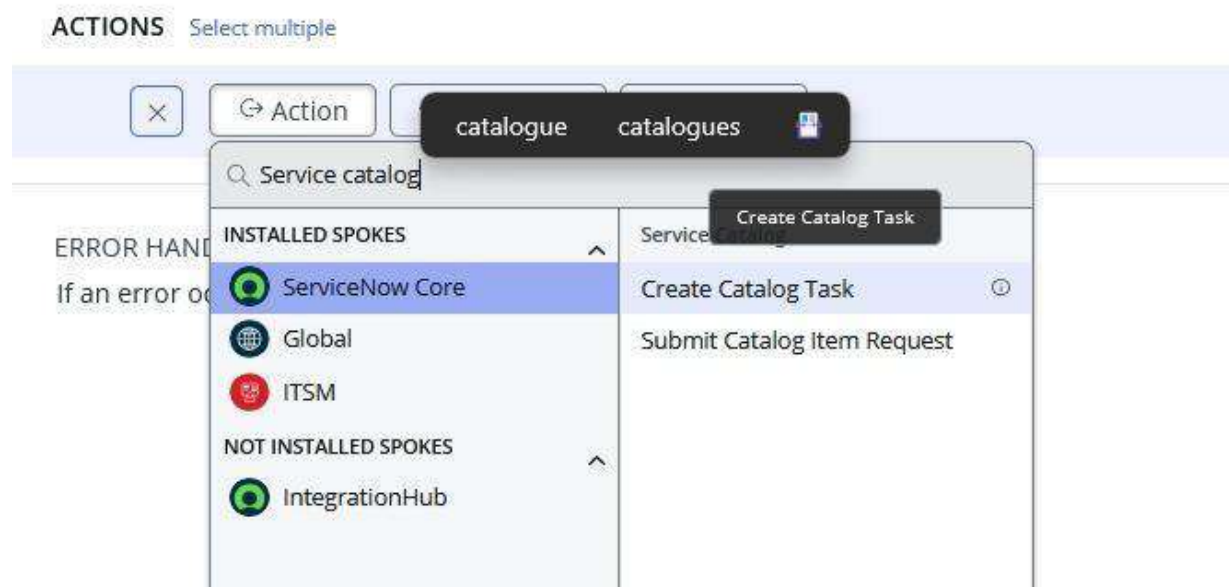
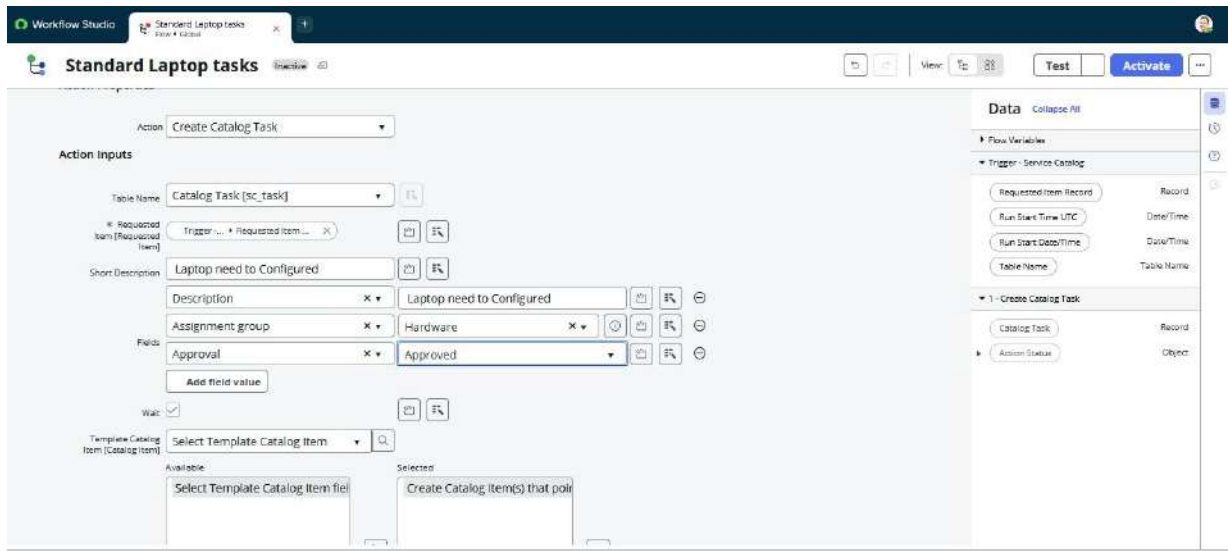


Figure 5.6: Configuring Catalog Task Fields

Short description,approve status,and assignment group are configured



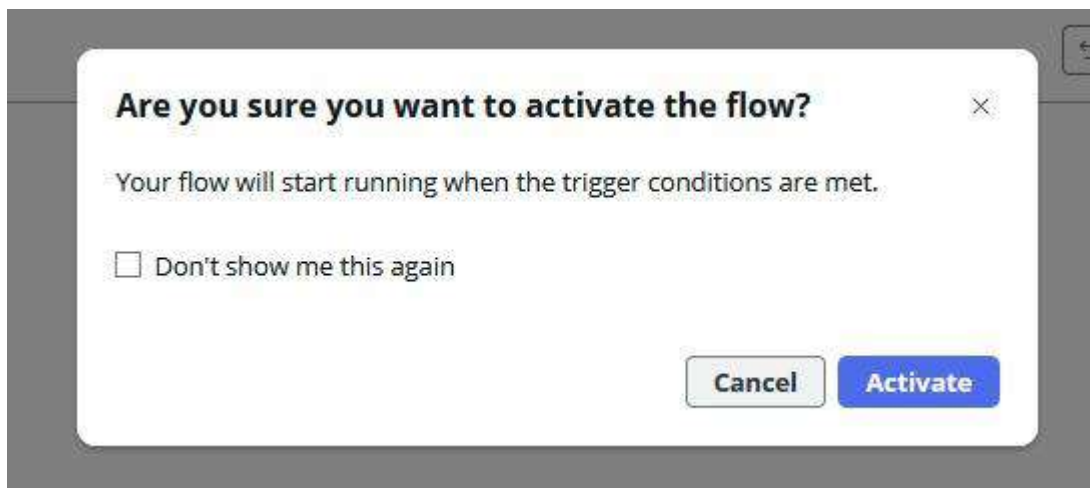
The screenshot shows the 'Standard Laptop tasks' workflow in Workflow Studio. The 'Action' is set to 'Create Catalog Task'. The configuration includes:

- Table Name:** Catalog Task [sc_task]
- Trigger:** Trigger... + Requested item...
- Short Description:** Laptop need to Configured
- Description:** Laptop need to Configured
- Assignment group:** Hardware
- Approval:** Approved
- Wait:** ☒
- Template Catalog Item:** Select Template Catalog Item

The right sidebar shows the 'Data' section with variables like 'Requested item Record', 'Run Start Time UTC', 'Run Start Date/Time', and 'Table Name'.

Figure 5.7: Activating the Flow

The completed flow is activated for execution.



The dialog box asks: "Are you sure you want to activate the flow?" and includes the following text:

Your flow will start running when the trigger conditions are met.

☐ Don't show me this again

Buttons: Cancel, Activate

Milestone 2 : Flow assignment

Figure 5.8: Opening Maintain Items

Maintain Items module is opened to assign the flow to a catalog item.



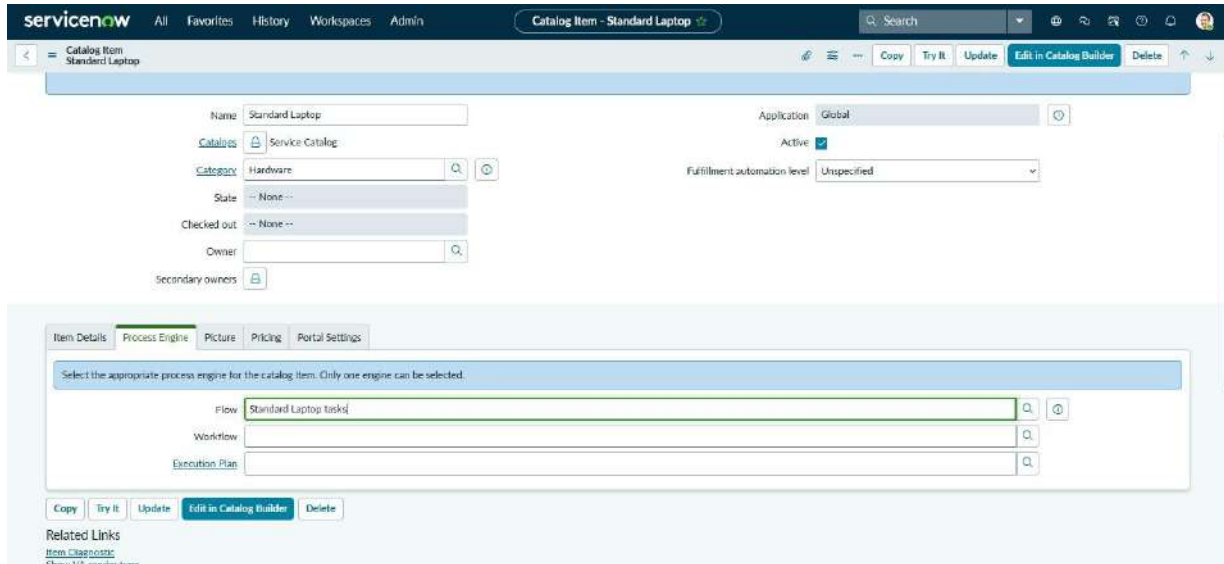
Figure 5.9: Selecting Standard Laptop Catalog Item

The Standard Laptop catalog item is selected for configuration.

servicenow All Favorites History Workspaces Admin Catalog Items									
Catalog Items									
Name	Short description	Active	Roles	Catalogs	Category	Price	Type	Updated	
Standard Laptop	Lenovo - Carbon x1	true		Service Catalog	Hardware	\$1,100.00	Item	2022-11-20 20:46:33	
StarTech Mini Display Port to DVI Adapter	PC / Mac Compatible	false		Service Catalog	Peripherals	\$17.23	Item	2022-11-20 20:46:33	
StarTech Mini Display Port to VGA Adapter	PC / Mac Compatible	true		Service Catalog	Peripherals	\$24.50	Item	2022-11-20 20:46:33	
StarTech USB Mini Hub	4 Port, USB 2.0	true		Service Catalog	Peripherals	\$11.55	Item	2022-11-20 20:46:33	
StarTech USB to DVI Adapter	Multi Monitor Adapter	true		Service Catalog	Peripherals	\$49.99	Item	2022-11-20 20:46:33	
STM MacBook Air Sleeve	STM Mac Sleeve - \$13"	true		Service Catalog	Peripherals	\$24.26	Item	2022-11-20 20:46:33	
STM MacBook Pro Sleeve	STM Mac Sleeve - \$15"	true		Service Catalog	Peripherals	\$25.78	Item	2022-11-20 20:46:33	
Table Index	Do you need to index a database table?	true		Service Catalog	Services	\$0.00	Item	2022-11-20 20:46:33	
Telephone Extension	IP 560 Phone	true		Service Catalog	Office	\$195.00	Item	2022-11-20 20:46:33	
	Request for temporary group				Application and			2022-05-01	

Figure 5.10: Assigning Flow to Catalog Item

The created flow is assigned to the Standard Laptop catalog item.



The screenshot shows the ServiceNow interface for configuring a catalog item. The top navigation bar includes 'servicenow', 'All', 'Favorites', 'History', 'Workspaces', and 'Admin'. The current page is 'Catalog Item - Standard Laptop'. The 'Catalog Item' section shows the following details:

- Name: Standard Laptop
- Application: Global
- Category: Hardware
- State: -- None --
- Checked out: -- None --
- Owner: [Search]
- Secondary owners: [Add]
- Active: ☒
- Fulfillment automation level: Unspecified

The 'Process Engine' tab is selected, showing a message: 'Select the appropriate process engine for the catalog item. Only one engine can be selected.' Below this, the 'Flow' field is set to 'Standard Laptop task'. The 'Workflow' and 'Execution Plan' fields are empty. At the bottom, there are buttons for 'Copy', 'Try It', 'Update', 'Edit in Catalog Builder', and 'Delete'. A 'Related Links' section is also visible at the bottom left.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Functional Testing

Test Case 1

Scenario: Create Flow

Expected Result: Flow should be saved and activated.

Actual Result: Success.

Test Case 2

Scenario: Assign Flow to Standard Laptop Catalog

Expected Result: Flow should be linked successfully.

Actual Result: Success.

Milestone 3 : Service Catalog

Test Case 3

Scenario: Place Laptop Order

Expected Result: Request should be created.

Actual Result: Success.

Figure 6.1: Standard Laptop Service Catalog Page



Service Catalog > Hardware > Standard Laptop

Lenovo - Carbon x1

x1 Carbon

The x1 Carbon is Lenovo's lightest ThinkPad yet. It provides a QHD display that fights glare and weighs less than three pounds. Ideal for most computing tasks, and highly mobile.

Technical Specs:

- Intel core i5 processor
- 512GB solid state drive (SSD)
- Backlit keyboard

Optional Software

☒ Adobe Acrobat

☐ Adobe Photoshop

Additional software requirements

Order this item

Price: \$1,100.00 + \$100.00 Annually

Quantity: 1

Subtotal: \$1,100.00 + \$100.00 Annually

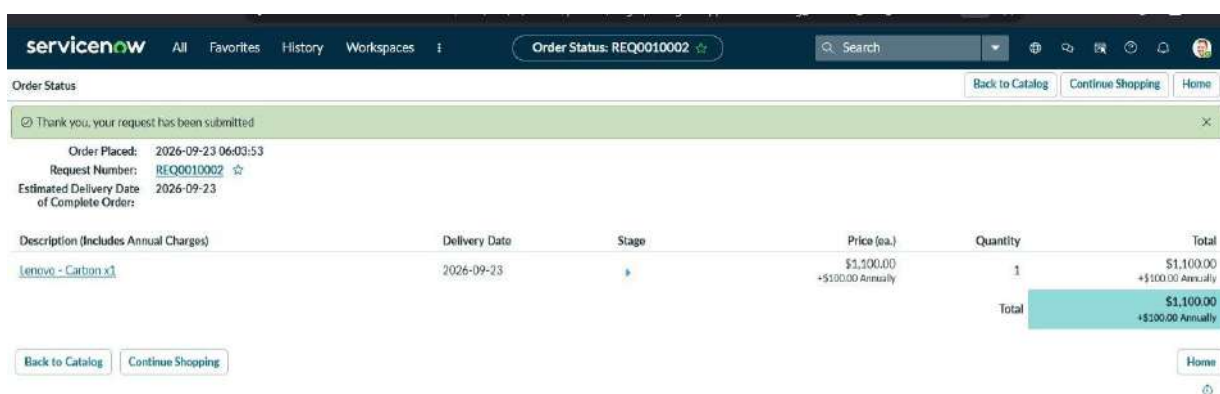
Delivery: [Order Now](#) 0 Days

[Order Now](#)

[Add to Cart](#)

Shopping Cart: Empty

Figure 6.2: Request Successfully Submitted



servicenow All Favorites History Workspaces Order Status: REQ0010002

Order Status

Thank you, your request has been submitted

Order Placed: 2026-09-23 06:03:53

Request Number: [REQ0010002](#)

Estimated Delivery Date of Complete Order: 2026-09-23

Description (Includes Annual Charges)	Delivery Date	Stage	Price (ea.)	Quantity	Total
Lenovo - Carbon x1	2026-09-23		\$1,100.00 +\$100.00 Annually	1	\$1,100.00 +\$100.00 Annually
Total					\$1,100.00 +\$100.00 Annually

[Back to Catalog](#) [Continue Shopping](#) [Home](#)

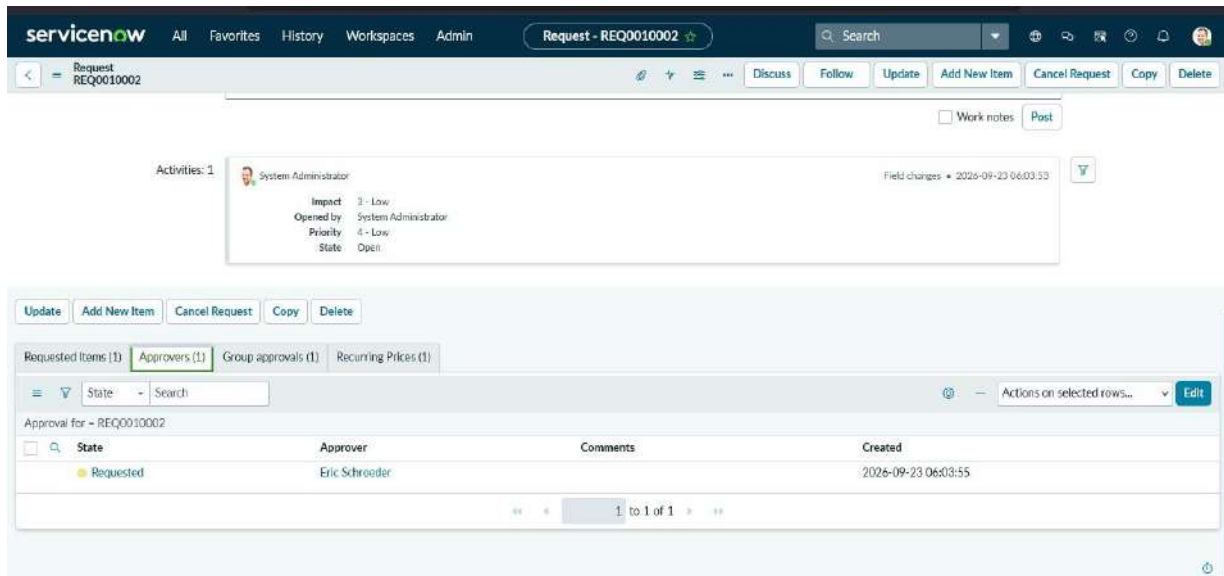
Test Case 4

Scenario: Approve Request

Expected Result: Catalog Task should be created automatically.

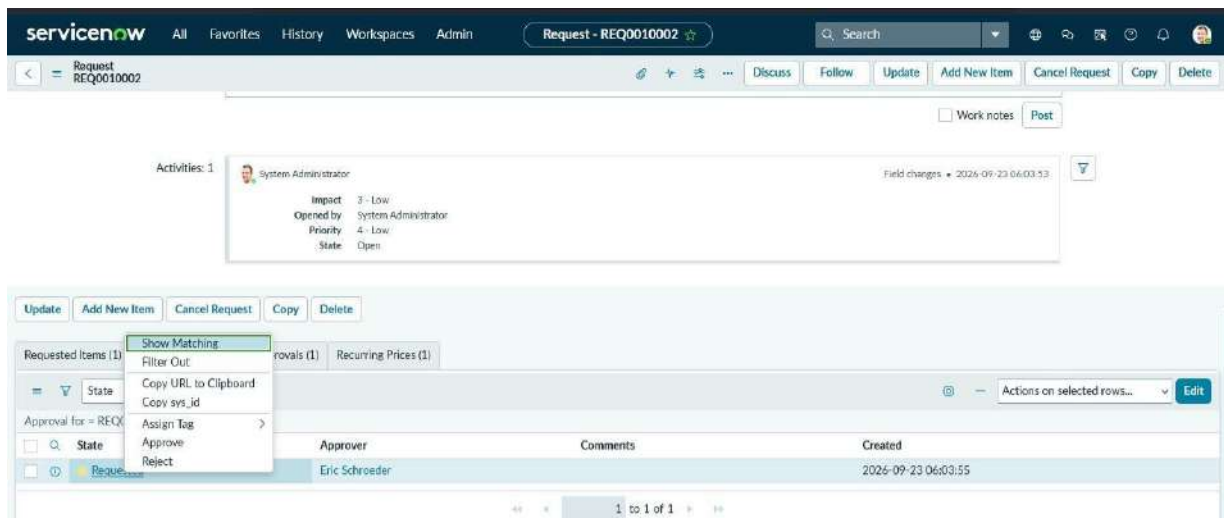
Actual Result: Success.

Figure 6.3: Request Record Showing Approver Details



The screenshot shows the ServiceNow interface for a Request record (REQ0010002). The top navigation bar includes 'servicenow', 'All', 'Favorites', 'History', 'Workspaces', and 'Admin'. The record title is 'Request - REQ0010002'. Below the title, there are tabs for 'Request', 'Activities', 'History', 'Workspaces', and 'Admin'. The 'Activities' tab is selected, showing a list of activities. The first activity is 'System Administrator' with a status of '3 - Low' and a priority of '4 - Low'. The activity details show 'Opened by: System Administrator', 'Priority: 4 - Low', and 'State: Open'. Below the activities, there are buttons for 'Update', 'Add New Item', 'Cancel Request', 'Copy', and 'Delete'. The 'Requested Items' tab is selected, showing a table with columns: 'State', 'Approver', 'Comments', and 'Created'. The table contains one row with the state 'Requested', approver 'Eric Schroeder', and creation date '2026-09-23 06:03:55'.

Figure 6.4: Approval of Service Request



The screenshot shows the ServiceNow interface for a Request record (REQ0010002). The top navigation bar includes 'servicenow', 'All', 'Favorites', 'History', 'Workspaces', and 'Admin'. The record title is 'Request - REQ0010002'. Below the title, there are tabs for 'Request', 'Activities', 'History', 'Workspaces', and 'Admin'. The 'Activities' tab is selected, showing a list of activities. The first activity is 'System Administrator' with a status of '3 - Low' and a priority of '4 - Low'. The activity details show 'Opened by: System Administrator', 'Priority: 4 - Low', and 'State: Open'. Below the activities, there are buttons for 'Update', 'Add New Item', 'Cancel Request', 'Copy', and 'Delete'. The 'Requested Items' tab is selected, showing a table with columns: 'State', 'Approver', 'Comments', and 'Created'. The table contains one row with the state 'Requested', approver 'Eric Schroeder', and creation date '2026-09-23 06:03:55'. A context menu is open over the 'Requested' state, showing options: 'Show Matching', 'Filter Out', 'Copy URL to Clipboard', 'Copy sys_id', 'Assign Tag', 'Approve', and 'Reject'.

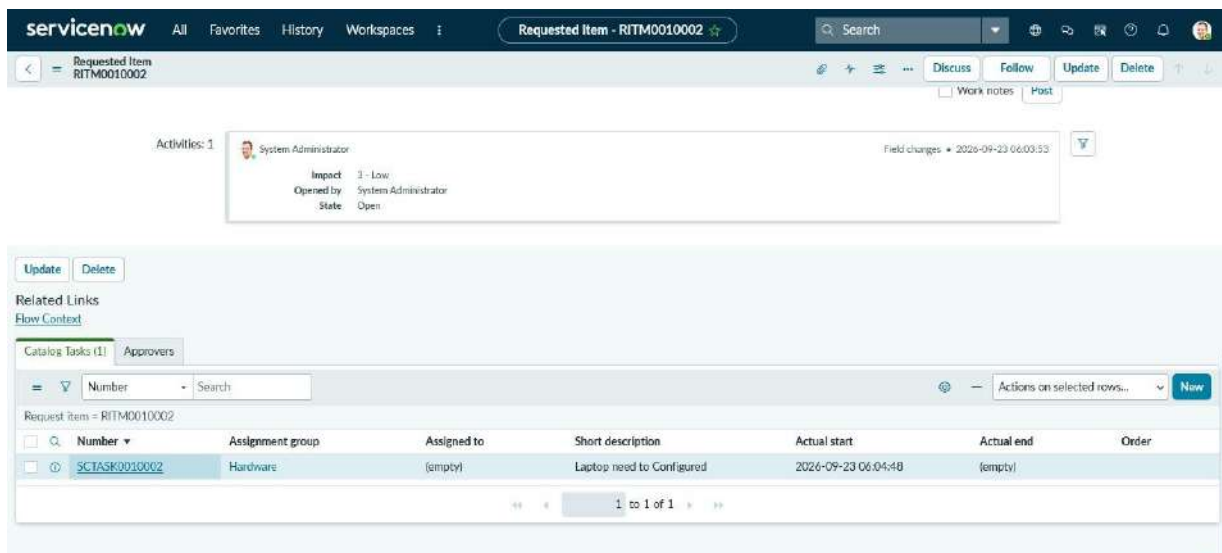
Test Case 5

Scenario: Verify Catalog Task

Expected Result: Short Description updated and Assignment Group set to Hardware.

Actual Result: Success.

Figure 6.5: Requested Item Record



servicenow All Favorites History Workspaces Requested Item - RITM0010002 Search

Requested Item
RITM0010002

Activities: 1

System Administrator

Impact: 3 - Low
Opened by: System Administrator
State: Open

Field changes: 2024-09-23 06:03:53

Update Delete

Related Links
[Flow Context](#)

Catalog Tasks (1) Approvers

Number Search

Request item = RITM0010002

Number	Assignment group	Assigned to	Short description	Actual start	Actual end	Order
SCTASK0010002	Hardware	(empty)	Laptop need to Configured	2024-09-23 06:04:48	(empty)	

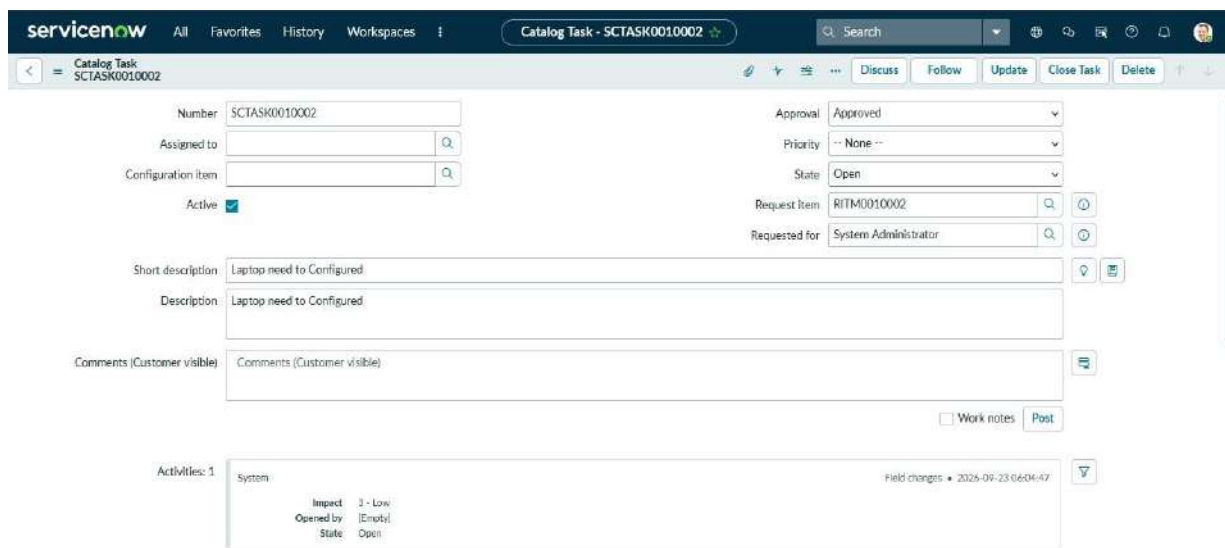
1 to 1 of 1

7. RESULTS

The automation successfully creates a Catalog Task after the approval of a Standard Laptop service request. The configured Flow Designer workflow executes automatically and performs the required actions without manual intervention. The Catalog Task is generated successfully, the assignment group is updated to the Hardware team, and the short description is modified according to the defined configuration. This demonstrates the effectiveness of workflow automation in streamlining IT procurement activities and improving operational efficiency.

Output Screenshots:

Figure 7.1: Catalog Task Automatically Created and Details



The screenshot displays the ServiceNow interface for a Catalog Task. The top navigation bar includes 'servicenow', 'All', 'Favorites', 'History', 'Workspaces', and a search bar. The main header shows 'Catalog Task - SCTASK0010002' with a search icon and a dropdown menu. Below the header, there are several tabs: 'Discuss', 'Follow', 'Update', 'Close Task', and 'Delete'. The form itself is divided into several sections. The 'Number' field is set to 'SCTASK0010002'. The 'Assigned to' field has a search icon. The 'Configuration item' field also has a search icon. The 'Active' checkbox is checked. The 'Approval' dropdown is set to 'Approved'. The 'Priority' dropdown is set to 'None'. The 'State' dropdown is set to 'Open'. The 'Request item' field is set to 'RITM0010002'. The 'Requested for' field is set to 'System Administrator'. The 'Short description' field contains 'Laptop need to Configured'. The 'Description' field also contains 'Laptop need to Configured'. The 'Comments (Customer visible)' field is empty. The 'Work notes' checkbox is unchecked. The 'Post' button is visible. The 'Activities' section shows a single activity named 'System' with a status of 'Open' and an impact of '3 - Low'. The 'Field changes' section shows a timestamp of '2025-09-23 06:04:47'.

8. ADVANTAGES & DISADVANTAGES

Advantages

- Reduces manual effort in processing service requests.
- Improves the speed and efficiency of IT procurement operations.
- Ensures consistent task creation and assignment.
- Minimizes the possibility of human errors.
- Enhances tracking and monitoring of request fulfillment activities.
- Provides a standardized workflow for laptop procurement requests.

Disadvantages

- Requires knowledge of ServiceNow configuration and Flow Designer.
 - Initial setup and configuration may require additional time.
 - Workflow changes require maintenance and testing.
 - Dependent on the proper functioning of ServiceNow components.
 - Requires knowledge of ServiceNow configuration and Flow Designer.
 - Initial setup and configuration may require additional time.
 - Workflow changes require maintenance and testing.
 - Dependent on the proper functioning of ServiceNow components.
 - Errors in flow configuration may affect the automation process.
-

9. CONCLUSION

The project "**Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer**" successfully demonstrates the use of ServiceNow Flow Designer to automate the laptop request fulfillment process. By automatically creating Catalog Tasks, updating task information, and assigning tasks to the appropriate Hardware support group, the solution eliminates manual activities and improves process efficiency. The implementation highlights the practical benefits of workflow automation in IT service management and contributes to a more reliable and streamlined procurement process.

10. FUTURE SCOPE

- Automatic email notifications to users and support teams.
 - SLA-based monitoring and escalation mechanisms.
 - Integration with Asset Management and Inventory Management modules.
 - Multi-level approval workflows for different request categories.
 - Automated asset allocation and tracking.
 - Dashboard and reporting capabilities for procurement monitoring.
 - Integration with third-party procurement and inventory systems.
-

11. APPENDIX

Source Code / Configuration

Flow Designer Configuration for Standard Laptop Request Automation.

Tools Used

- ServiceNow Platform
- Flow Designer
- Service Catalog
- Catalog Task Management

Dataset Link

- Not Applicable.

GitHub / Project Demo Link

- **GitHub Link :** <https://github.com/Selvapushpam07/Streamlining-IT-Procurement-Automating-Standard-Laptop-Orders-with-Flow-Designer>
- **Project Demo Link :** <https://docs.google.com/videos/d/16LYzrxWMThTCtNLZYK1lsIKh4boFv7pojtZ8yI9OPFQ/play?usp=sharing>

References

- ServiceNow Documentation
 - ServiceNow Flow Designer User Guide
 - Service Catalog Administration Guide
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